

PLSC 197 / SODA 110N: Social Data, Technology, and Artificial Intelligence

Applied Lab One: Data Visualization
Part II: AI-Based Visualization Tools

Lab

Reminer: The goal for this week-long-lab is to familiarize you with some of the no-code tools available for visualizing data. This lab will be assigned in three parts: A Monday-to-Wednesday assignment (*Part I*), a Wednesday-to-Friday assignment (*Part II*), and a third part due next week (*Part III*). This is *Part II*.

The entire lab will rely on the 2019 subset of the [World Development Indicators](#) (WDI) data that we used as our example data in class. Those data have ≈ 40 variables on roughly 215 countries in the world, with values measured in 2019. The description of those data is [here](#), and the data itself is [here](#), in `.csv` format.

Part II of the lab assignment is almost as simple (or perhaps even simpler) than Part I:

1. Open up a session in Penn State's [AI Studio](#) and log in.
2. Choose a model (one of OpenAI's ChatGPT, Anthropic's Claude, or Google's Gemini).
3. Start with a prompt indicating that you're uploading the [WDI data](#). You can upload the data using the "+" button in the prompt window, and/or point the model to those data on the web at the Github URL.
4. Once again, choose *one* continuous variable, and *one* discrete variable. Pick ones that are of interest to you; be sure not to choose the same variable for both. You are welcome to use the ones you chose for Part I.
5. Use the app to create visual summaries / plots of:
 - (a) The continuous variable,
 - (b) The discrete variable,
 - (c) The two variables together.¹

Keep track of the specific prompts that you use. This is our first collective foray into the messy world of "prompt engineering," so it's good to have a record of what works and what doesn't.

Details

You should complete (or, at least, make progress toward completing) this part of the lab before Friday's class (September 18, 2026), and come to class on that day prepared to discuss your preliminary work and ask questions. Once again, you do not have to submit anything prior to Friday's class. The entire lab exercise is mandatory, and is worth a possible 150 points. **Points will be assessed upon completion of all three parts, the last of which will be due on Friday, September 25, 2026.**

¹Once again, there are a few ways to do this. In this case, none of them depend on the app, which is nice(r).